1. A gear manufacturer operates two production sites in Halifax and Calgary and produces three gear types: A, B, and C. The Halifax site is a newer plant with state-of-the-art equipment but fewer experienced workers. This results in differences in production costs and the production rates (units/hour) as below:

	Production Cost			Production Rate per Hour			Production available
	Gear A	Gear B	Gear C	Gear A	Gear B	Gear C	
Halifax	\$0.60	\$1.00	\$0.32	3,000	1,000	8,000	170 hours
Calgary	\$0.50	\$0.90	\$0.35	2,500	800	6,000	150 hours

The selling prices are \$2.3 for gear A, \$2.2 for gear B, and \$1.1 for gear C. What should they make and where to maximize profits?

Decision Variables:

AH: # of gear A made in Halifax
BH: # of gear B made in Halifax
CH: # of gear C made in Halifax
AC: # of gear A made in Calgary
BC: # of gear B made in Calgary
CC: # of gear C made in Calgary

Objective function:

Profit $= (2.3 - 0.6)AH + (2.2 - 1)BH + (1.1 - 0.32)CH + (2.3 - 0.5)AC + (2.2 - 0.9)BC + (1.1 - 0.35)CC$
$= (1.7)AH + (1.2)BH + (0.78)CH + (1.8)AC + (1.3)BC + (0.75)\,CC$

Halifax can produce 3,000 gear A per hour. This means that producing ONE gear A takes $\frac{1}{3{,}000}$ hour. Similarly, it takes $\frac{1}{1{,}000}$ to produce one gear B and $\frac{1}{8{,}000}$ to produce one gear C in Halifax. If we produce AH gear A in Halifax, since each takes $\frac{1}{3{,}000}$ hour, then $AH\ (\frac{1}{3{,}000})$ is total hours required by gear A in Halifax. Alternatively, we can write $AH\ (\frac{1}{3{,}000})$ as $\frac{AH}{3{,}000}$.

Now, we know that total hours used for production in Halifax cannot exceed 170 hours, i.e.:

$$\frac{AH}{3{,}000} + \frac{BH}{1{,}000} + \frac{CH}{8{,}000} \leq 170$$

We can write a similar constraint for Calgary. The LP model is:

$$\textbf{Maximize}\ \ 1.7AH + 1.2BH + 0.78CH + 1.8AC + 1.3BC + 0.75CC$$

$$\textbf{Subject to:}\quad \frac{AH}{3{,}000} + \frac{BH}{1{,}000} + \frac{CH}{8{,}000} \leq 170$$

$$\frac{AC}{2{,}500} + \frac{BC}{800} + \frac{CC}{6{,}000} \leq 150$$

$$AH, BH, CH, AC, BC, CC \geq 0$$

2. A retired couple is willing to invest \$750,000 in the listed companies. Although all investments appear safe, they do not want to invest more than 25% of their money in any of them. They also want to invest at least half of the money in long-term bonds that mature in 10 or more years. Finally, they want no more than 35% of the funds to go to companies with a low rating (3 or 4). What is the amount of investment in each company?

Company	Rate of return	Years to Maturity	Rating
1	8.65%	11	1-Excellent
2	9.50%	10	3-Good
3	10.00%	6	4-Fair
4	8.75%	10	1-Excellent
5	9.25%	7	3-Good
6	9.00%	13	2-Very Good

Decision Variables: Let x_i be the amount of investment in the company i , where $i = 1,2,3,4,5,6$

Then, the LP model is:

$$Max\ 0.0865x_1 + 0.095x_2 + 0.1x_3 + 0.0875x_4 + 0.0925x_5 + 0.09x_6$$

$S.t.$

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \leq 750{,}000$$

$$x_i \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6), \quad \text{for all } i = 1,2,3,4,5,6$$

$$x_1 + x_2 + x_4 + x_6 \geq 0.5\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_2 + x_3 + x_5 \leq 0.35\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_i \geq 0, \quad \text{for all } i = 1,2,3,4,5,6$$

This is in fact 6 constraints:

$$x_1 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_2 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_3 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_4 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_5 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

$$x_6 \leq 0.25\ (x_1 + x_2 + x_3 + x_4 + x_5 + x_6)$$

3. A company is planning its production for X over the next 6 months. The warehouse capacity is 600, and the company is willing to hold 150 of X as safety stock. To maintain a stable workforce, the company aims to produce at least half of its maximum monthly capacity. The company uses ending inventory as the number of units carried (held) each month. The current inventory is 275. Find the optimal (minimum cost) production/Inventory plan.

Product	X					
Month	1	2	3	4	5	6
Demand	100	450	600	550	350	400
Production Cost	\$240	\$250	\$265	\$285	\$280	\$260
Inventory Holding Cost	\$3.60	\$3.75	\$3.98	\$4.28	\$4.20	\$3.90
Max Production Capacity	400	350	400	450	400	350

Inventory Balancing formula: $I_t = I_{t-1} + P_t - D_t$

Decision Variables:

P_i is the amount of production for X in the month i, where $i = 1,2,3,4,5,6$

I_i is the amount of inventory for X at the end of the month i , where $i = 1,2,3,4,5,6$

Objective Function:

$$Min\ 240P_1 + 250P_2 + 265P_3 + 285P_4 + 280P_5 + 260P_6$$
$$+3.6I_1 + 3.75I_2 + 3.98I_3 + 4.28I_4 + 4.2I_5 + 3.9I_6$$

Production Max Constraints:

$P_1 \leq 400,\quad P_2 \leq 350,\quad P_3 \leq 400,\quad P_4 \leq 450,\quad P_5 \leq 400,\quad P_6 \leq 350$

Production Min Constraints:

$P_1 \geq 200,\quad P_2 \geq 175,\quad P_3 \geq 200,\quad P_4 \geq 225,\quad P_5 \geq 200,\quad P_6 \geq 175$

Inventory Balance Constraints for X:

$$P_1 - I_1 = 100 - 275 = -175$$
$$P_2 + I_1 - I_2 = 450$$
$$P_3 + I_2 - I_3 = 600$$
$$P_4 + I_3 - I_4 = 550$$
$$P_5 + I_4 - I_5 = 350$$
$$P_6 + I_5 - I_6 = 400$$

Inventory Max Constraints:

$I_i \leq 600, i = 1,2,\ldots,6$

Inventory Min Constraints:

$I_i \geq 150, i = 1,2, \dots ,6$

Non-negativity constraints:
$P_i, I_i \geq 0, \quad \text{for all } i = 1,2,3,4,5,6$